ParenTalk: LLM-Based Communication Training for Parents in Children's Academic Support

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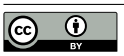
Abstract

Parent-child communication plays a fundamental role in children's academic and developmental outcomes, yet in the Chinese educational context, parents' heavy involvement combined with limited communication competency often produces frustration and conflict rather than support. Informed by interviews with 10 parents, we designed ParenTalk, a three-stage system powered by LLMs and grounded in Satir's communication model. ParenTalk is designed to help parents reflect on and practice communication with their children through interactive, personalized exercises. Preliminary evaluation with 40 parents suggest that ParenTalk offers a practical and immersive learning experience, effectively improving parents' communication skills and family relationships.

CCS Concepts

• **Do Not Use This Code → Generate the Correct Terms for Your Paper**; *Generate the Correct Terms for Your Paper*; Generate the Correct Terms for Your Paper; Generate the Correct Terms for Your Paper.

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1 Introduction

Effective parent-child communication during academic support activities is important for children's learning outcomes and long-term motivation [4–6]. Yet many parents lack the communication skills needed to engage their children constructively around schoolwork, which creates most of the parent-child conflicts [3]. This issue is especially salient in China, where parents often serve as primary academic supporters [8]. Traditional approaches to improving academic parent-child communication, such as family therapy [7] or behavioral parent training [1], face scalability issues due to time and cost limitations [2]. Technology-facilitated interventions have sought to lower these barriers, but most deliver static, one-size-fits-all content that does not adjust to individual family needs or specific challenges faced by different parent-child dyads [1].

Recent work in HCI has explored large language models (LLMs) and conversational agents (CAs) for communication skill training, showing that simulated practice with feedback can measurably improve communication skills [10]. However, these efforts primarily target interpersonal conflict [12] or family topics like storytelling [13]. Few studies have examined the communication challenges parents face within the academic-support settings. Moreover, the potential of LLMs to address these challenges remains underexplored:

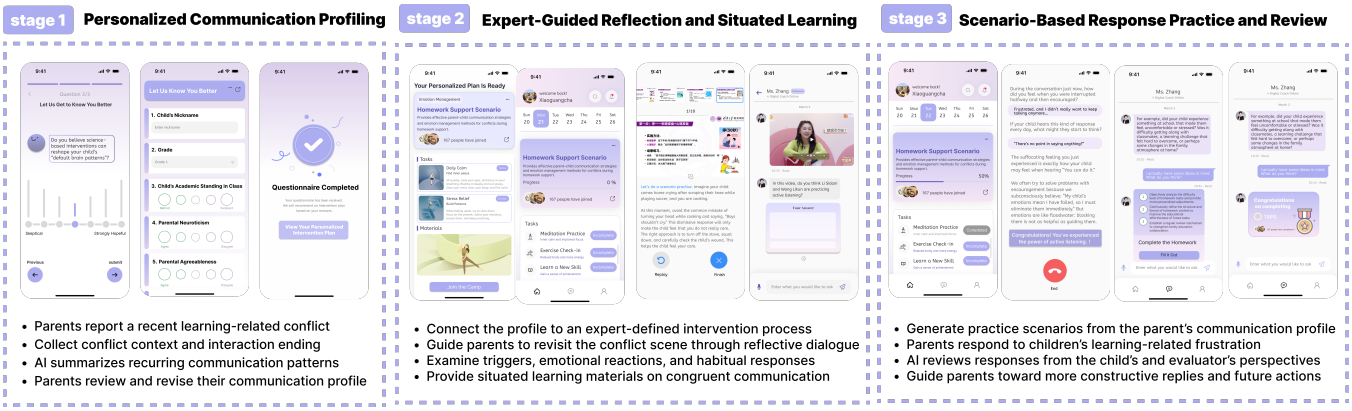


Figure 1: Overview of ParenTalk's three-stage workflow. The system helps parents identify communication patterns from a recent learning-related conflict, engage in expert-guided reflection and situated learning, and practice revised responses through AI-generated scenarios with feedback and review.

most existing systems offer little more than an open conversational interface [9], without a systematic intervention pipeline that assesses each parent-child conflict, guides structured learning, and delivers tailored feedback.

To address this, we designed ParenTalk, an LLM-powered mobile application that helps parents reflect on and practice communication skills for supporting their children's schoolwork. The system structures the learning experience as a guided, three-stage pipeline grounded in Satir's communication model [11]: It (1) identifies the parent's communication patterns by surfacing problematic behaviors in their academic-support interactions, (2) delivers structured intervention and situated learning about congruent communication, and (3) engage parents in practicing responses within AI-simulated dialogues, providing turn-by-turn feedback and a post-session review that offers concrete suggestions for improvement.

2 Formative Study

We conducted formative interviews with 10 Chinese parents of children aged 6–16, who had difficulty communicating with their children about academic topics and had used AI or other tools to support their communication. Participants were recruited through Chinese social media and compensated 30 RMB for a 30-minute interview, with school ethics approval. From the results, we derive four design goals:

DG1. Ground the system's guidance in expert-validated knowledge and established parent-child communication theory.

DG2. Support parents in practicing communication skills through realistic, scenario-based exercises that reinforce learning.

DG3. Provide a highly personalized learning environment that helps parents address their specific challenges in ways they prefer.

DG4. Foster parents' self-awareness and reflection to encourage lasting behavior change.

3 System Design and Implementation

Based on our design goals, ParenTalk guides parents through three stages (Fig. 1). A chat interface is also retained for parents to ask questions on demand. ParenTalk uses GPT-4o to support conflict summarization, reflective question generation, response feedback, and session review, while structured task flows constrain how these LLM functions are used in each stage.

Stage 1: Personalized Communication Profiling (DG3). Parents first answer guided questions about a recent learning-related conflict. The system records the conflict context, the parent's emotional reaction, the child's response, and how the interaction ended. Based on these inputs, the AI summarizes the communication pattern shown in the conflict and generates a profile that parents can review and revise before continuing to later tasks.

Stage 2: AI-Guided Reflection and Learning (DG1, DG3, DG4). Using the profile from Stage 1, the system connects the identified pattern to an expert-defined intervention process grounded in Virginia Satir's communication model and parent-child communication guidance. The AI uses reflective questions and scenario-based voice dialogue to bring parents back to the conflict scene and help them examine the triggers, emotional reactions, and recurring responses behind the interaction. The system also provides brief learning materials validated by educational experts, such as videos or slides, to introduce congruent communication and strategies for emotional validation and conflict repair.

Stage 3: Scenario-Based Response Practice and Review (DG2). Parents then take part in scenario-based practice generated from their communication profile. In each scenario, they respond to a child's learning-related frustration and can revisit responses that may sound blaming, dismissive, or hard to use in the moment. The AI reviews the response from both the child's perspective and an evaluator's perspective, then help parents revise the response in their own words. At the end of the session, the AI generates a summary covering the pattern identified, practiced strategies, and possible actions for future learning-related conflicts.

4 Preliminary Field Evaluation

We deployed ParenTalk in the field with 40 parents who need support with academic communication. Stage 1 assessed each parent's communication issues and generated a personalized 7-day

camp from a library of 14 expert-designed modules spanning self-awareness, congruent communication, and emotion management. Each day's 30 minute module guided parents through Stage 2 and 3. Daily surveys and a post-camp survey measured satisfaction, self-reflection, skill gains, relationship change, and usability, with 10 parents (P1–P10) completing follow-up interviews. Participants were compensated 100 RMB and all procedures were approved by the authors' university.

Perceptions of the AI-Guided Learning Format. Parents largely preferred the AI-guided learning format to plain video or open-ended AI chatbots, arguing that having to read, think, and respond made lessons stick better under limited daily attention (P1, P2, P6, P7). Several found videos better at prompting reflection, but easier to zone out on, and valued ParenTalk's interleaving of short videos within the AI dialogue (P1, P4, P9).

Effectiveness. Parents found the recommended lessons well-targeted (P1, P3) and the strategies practical (P3, P5, P8), most reporting gains in communication and emotion-regulation skills. However, several noted a lack of review of earlier content, making strategies that felt clear right after a session fade within days, particularly for older parents (P5, P10).

Reflectiveness. Parents valued that ParenTalk turned attention to their own behavior rather than the child's, and described this self-reflection as a precondition for behavior change (P2, P3, P5). Most reported new awareness of their own patterns after the camp and started changing.

Usage Patterns. Parents spent 20–30 minutes daily on camp, mostly during commutes, work downtime, or while their child did homework, and were satisfied with this pace. They also opened the AI chat for just-in-time questions, e.g., why their child had just reacted angrily (P3, P8). Besides, parents felt 7 days was too short for lasting change and wanted a long-term, continuously adaptive program (P1, P2, P4, P5, P6, P7).

5 Conclusion and Future work

We presented the design of ParenTalk, an LLM-powered parent-training tool that helps parents reflect on and improve their communication with children. Initial user study with 40 parents suggests that ParenTalk offers a practical and immersive learning experience, effectively improving parents' communication skills and family relationships. Future work includes recruiting a broader participant pool and quantitatively investigating how specific system features influence users' learning experience and outcomes. Besides, children feedback should be included in future studies to reflect communication improvements and other challenges.

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